SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	K.Nilavan	**Reg. No.:**	192511028
Section / Batch:	2025-2029	**Date:**	28/08/2026

Code of Conduct

I K.Nilavan (192511028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Nilavan

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

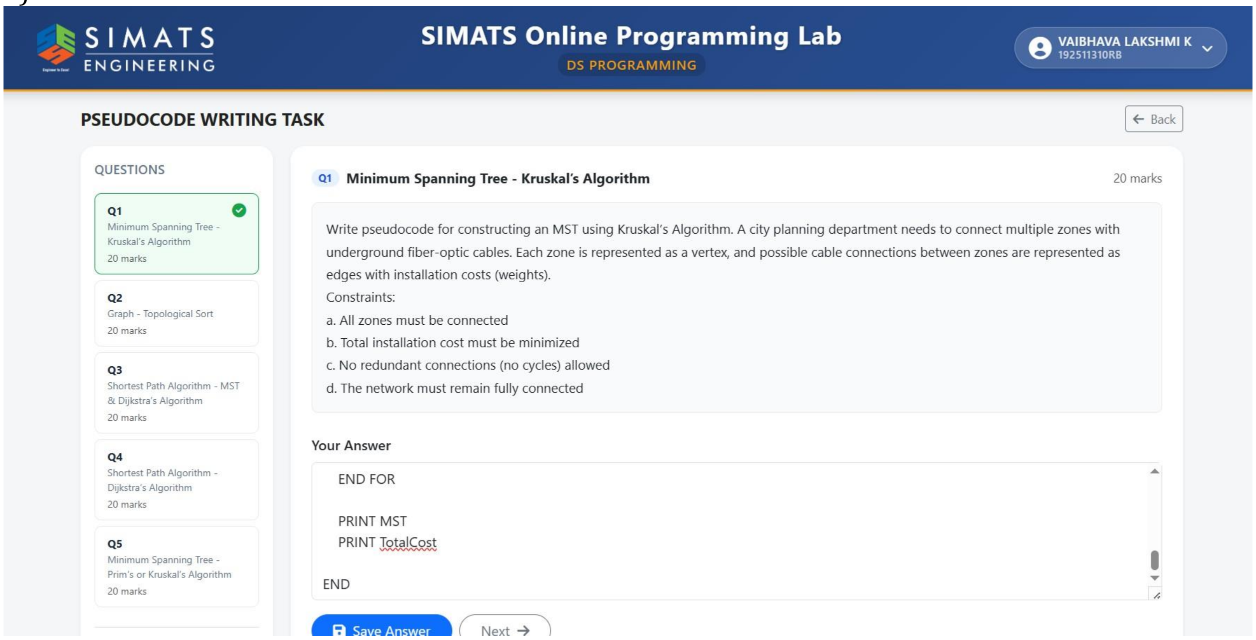
Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

1)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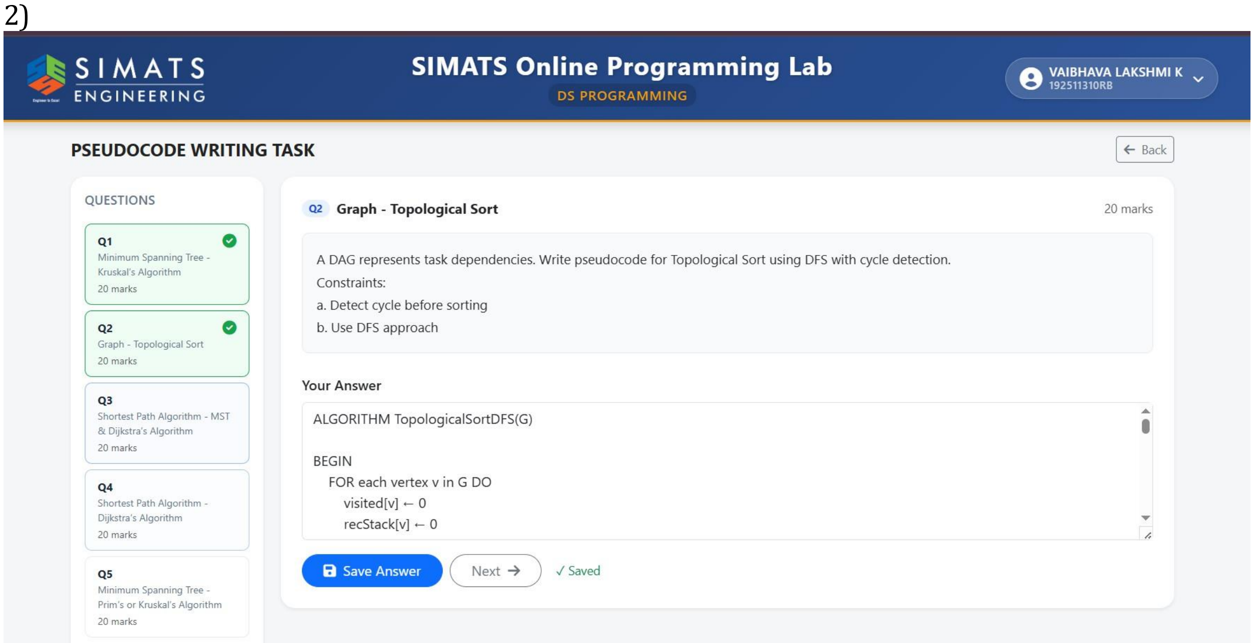
QUESTIONS
Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Q2
Graph - Topological Sort
20 marks
Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks
Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
Q1 Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected
Your Answer
END FOR
PRINT MST
PRINT TotalCost
END
Save Answer
Next

2)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Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
Q2 Graph - Topological Sort
20 marks
A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach
Your Answer
ALGORITHM TopologicalSortDFS(G)
BEGIN
FOR each vertex v in G DO
visited[v] ← 0
recStack[v] ← 0
Save Answer
Next
✓ Saved

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QUESTIONS

Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm 20 marks

Q3 **Shortest Path Algorithm - MST & Dijkstra's Algorithm** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
ALGORITHM MST_Dijkstra(G, Source, Destination)

BEGIN

   // Step 1: Construct MST for infrastructure
   MST ← empty set
```

Save Answer   Next →

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QUESTIONS

Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
ALGORITHM DijkstraWithTollFreeVoucher(G, S, D)

BEGIN

   FOR each vertex v DO
      distance[v][0] ← ∞
```

Save Answer   Next →

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QUESTIONS

Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Q2 Graph - Topological Sort 20 marks

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm 20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm 20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

Q5 **Minimum Spanning Tree - Prim's or Kruskal's Algorithm** 20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
ALGORITHM SecondBestMST(G)

BEGIN

   // Step 1: Find the Minimum Spanning Tree
   MST ← Kruskal(G)
```

Save Answer   ✓ Saved